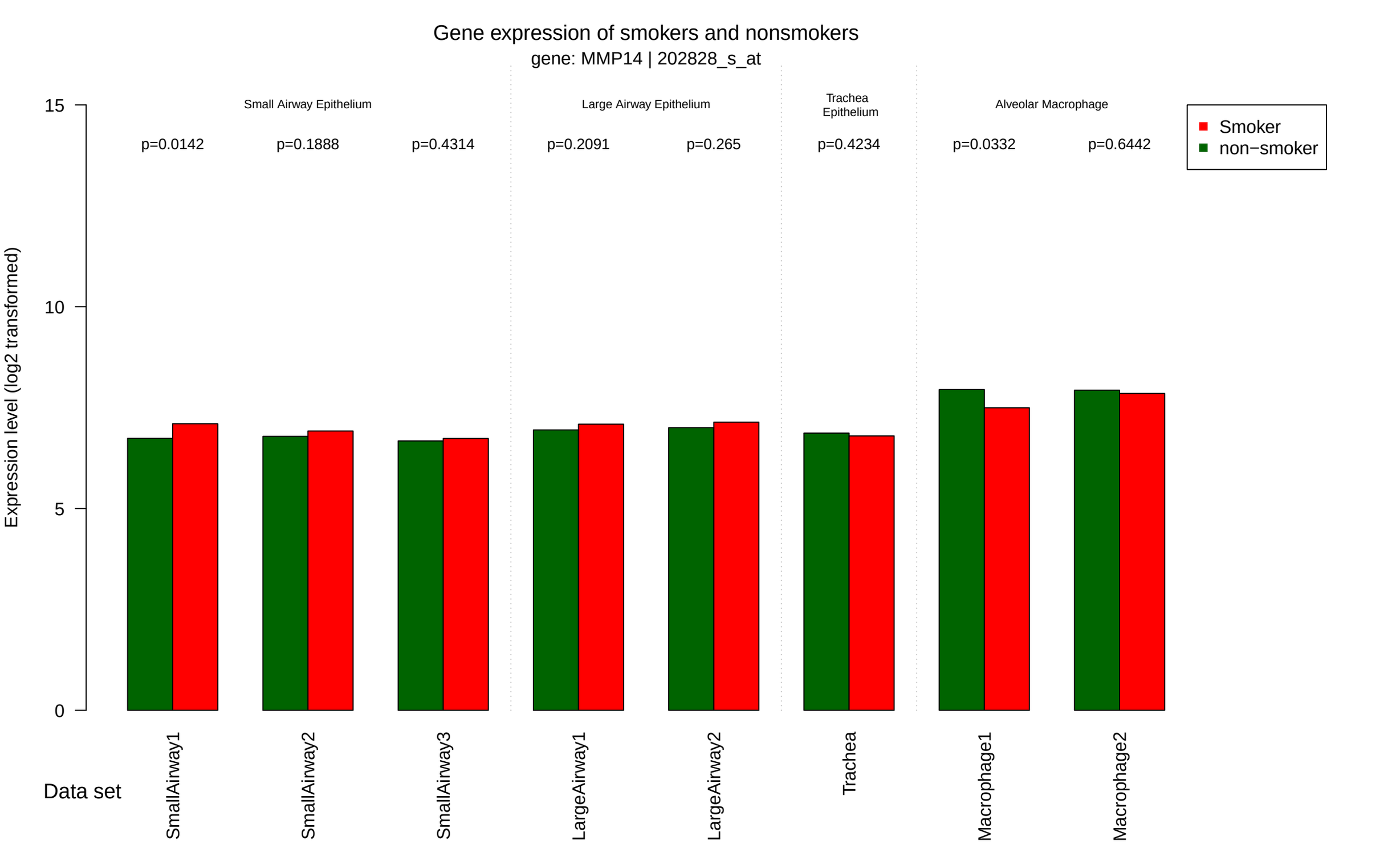
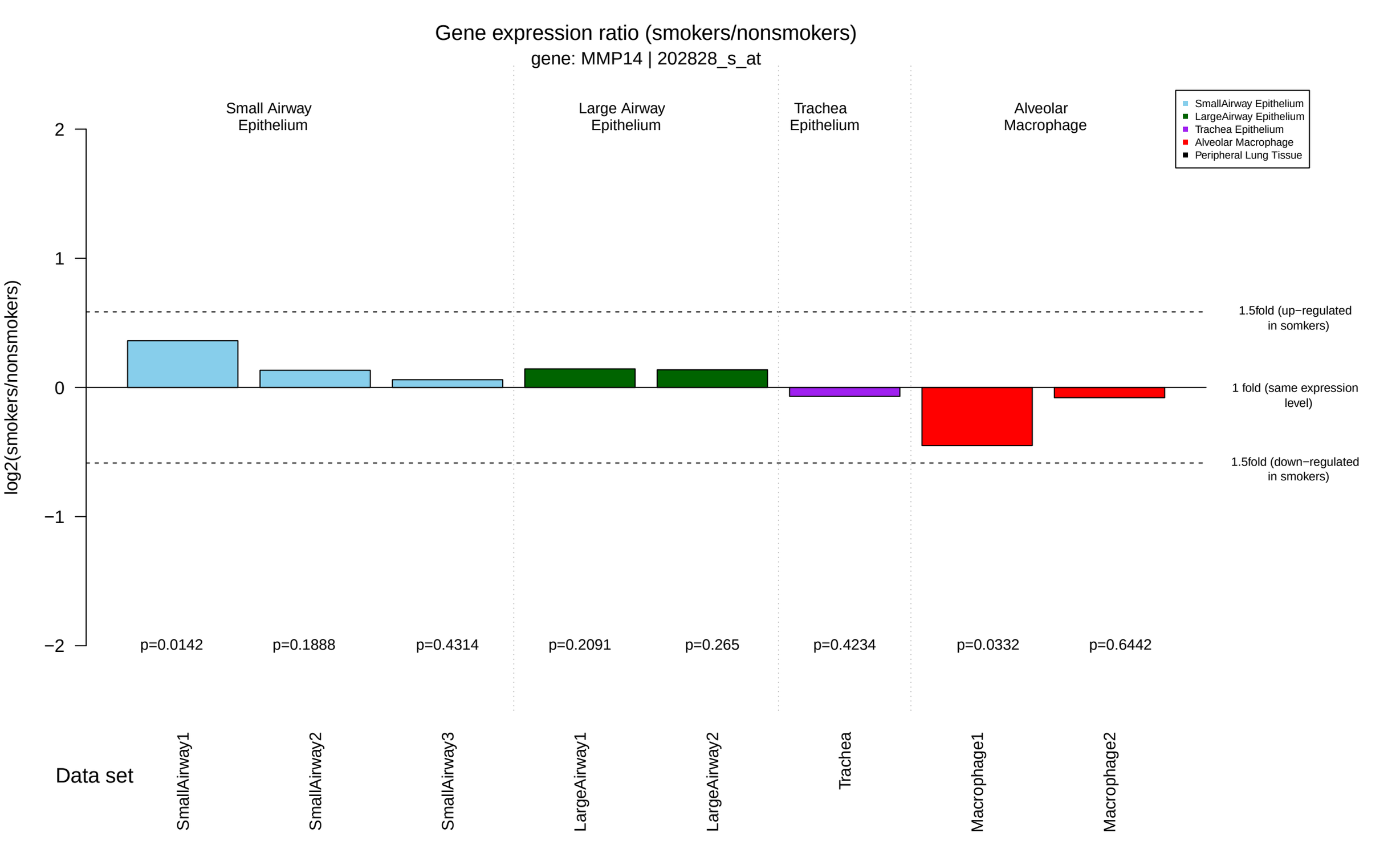
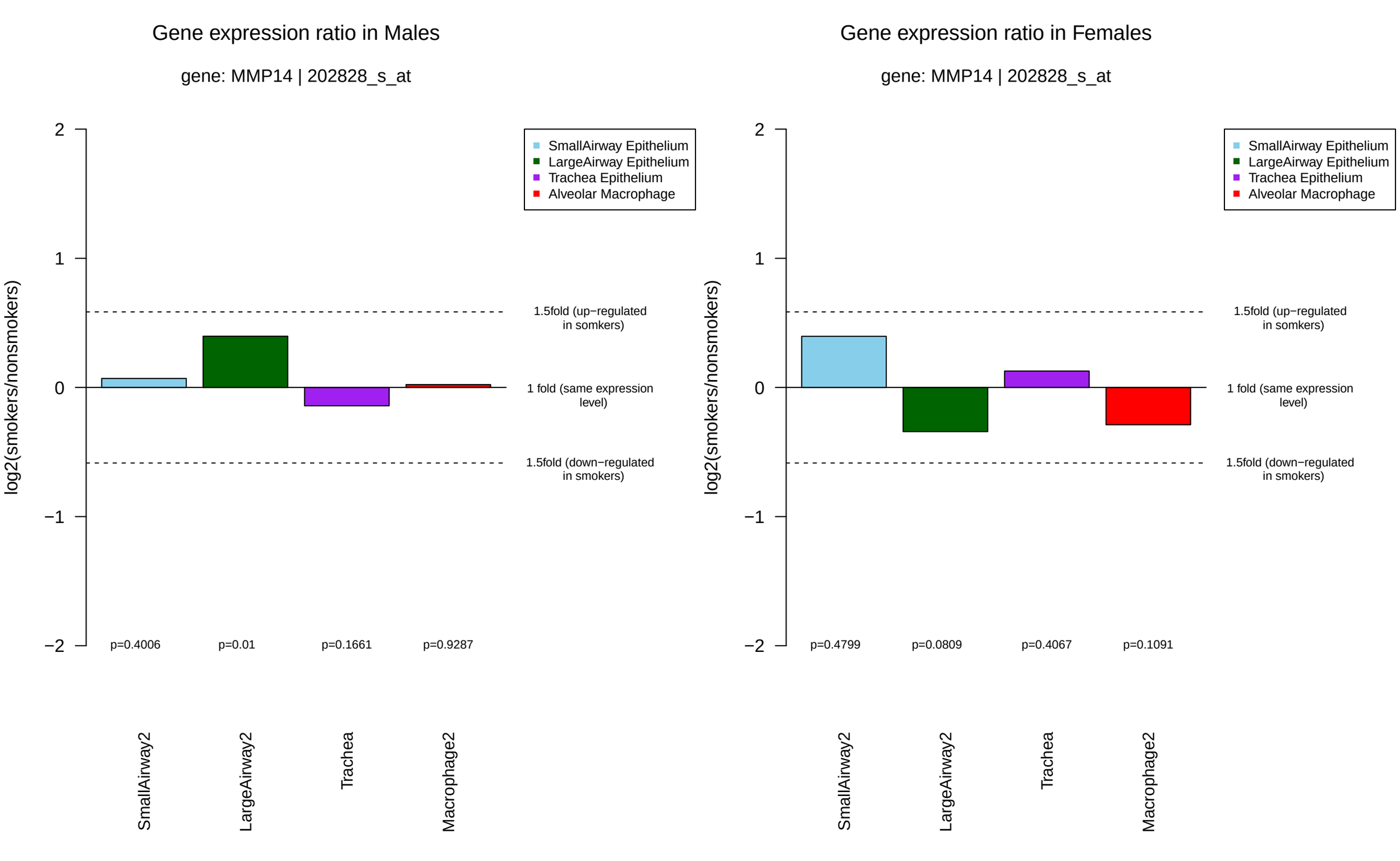
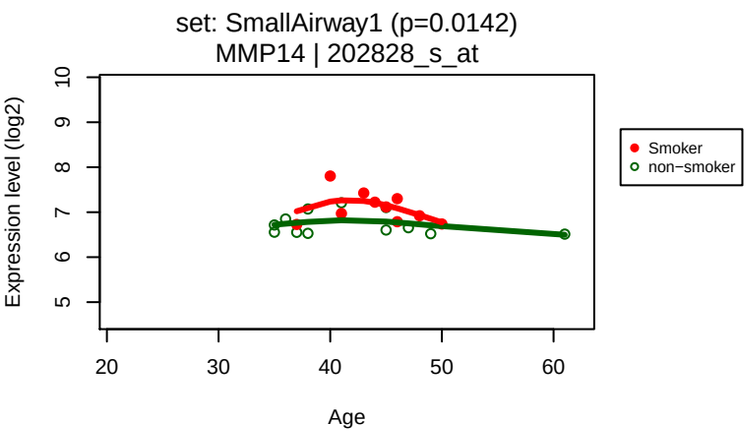
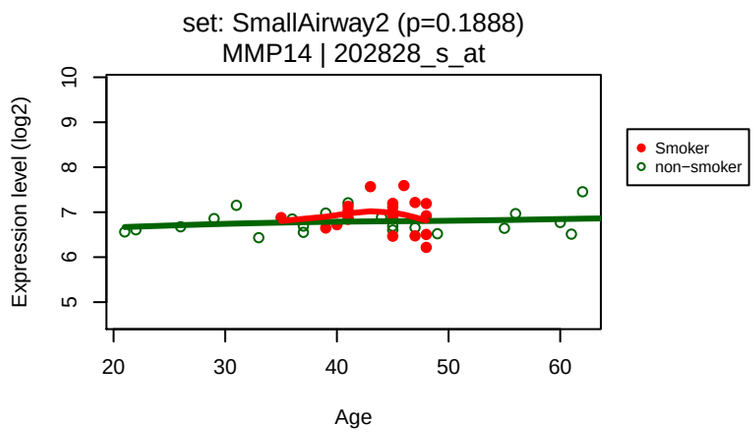
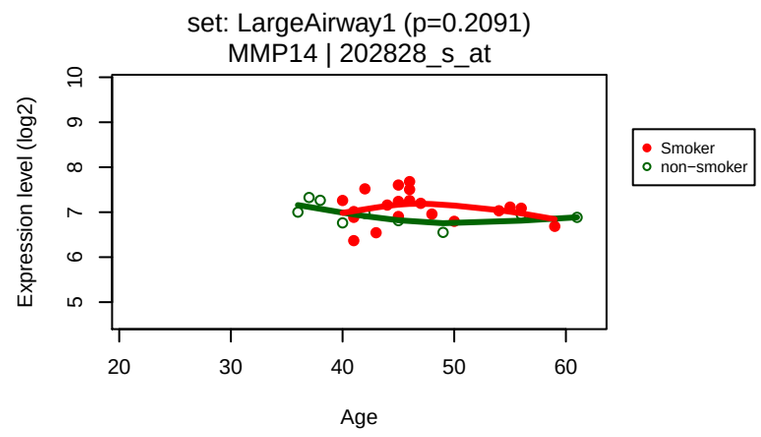

Gene expression of smokers and nonsmokers
gene: MMP14 | 202828_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0142
p=0.1888
p=0.4314
p=0.2091
p=0.265
p=0.4234
p=0.0332
p=0.6442
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

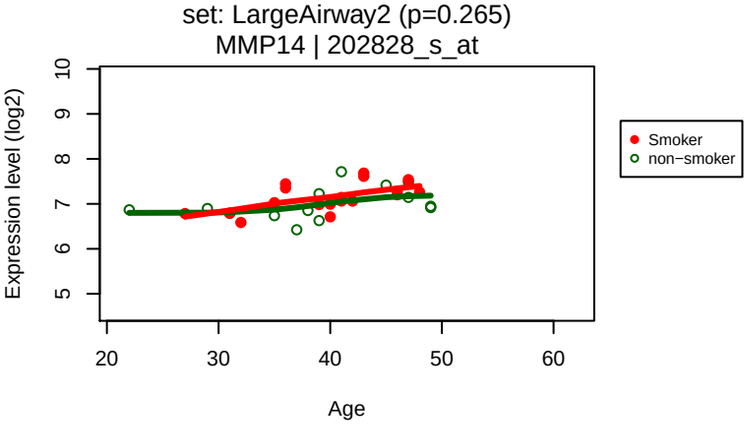

Gene expression ratio (smokers/nonsmokers)
gene: MMP14 | 202828_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0142
p=0.1888
p=0.4314
p=0.2091
p=0.265
p=0.4234
p=0.0332
p=0.6442
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

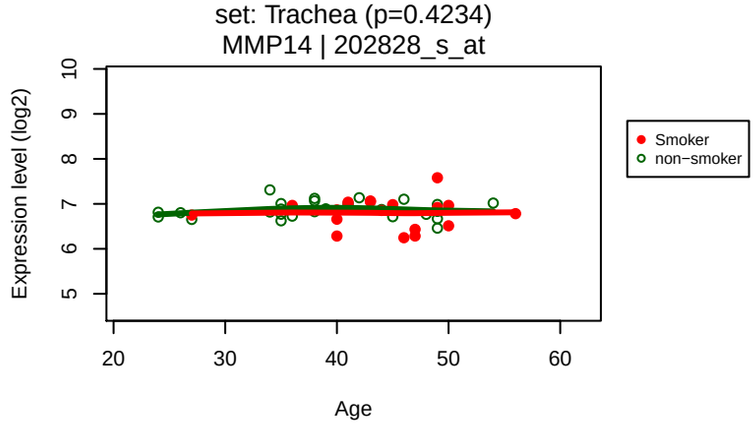

Gene expression ratio in Males
gene: MMP14 | 202828_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4006
p=0.01
p=0.1661
p=0.9287
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP14 | 202828_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4799
p=0.0809
p=0.4067
p=0.1091
SmallAirway2
LargeAirway2
Trachea
Macrophage2

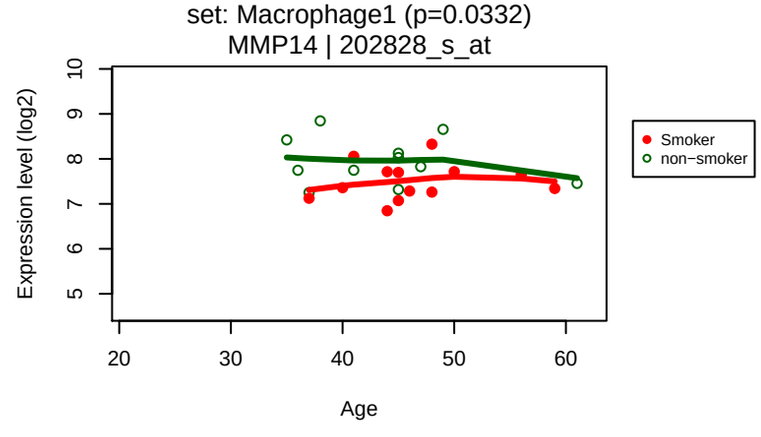

set: SmallAirway1 (p=0.0142)
MMP14 | 202828_s_at
Expression level (log2)
Age
Smoker
non−smoker

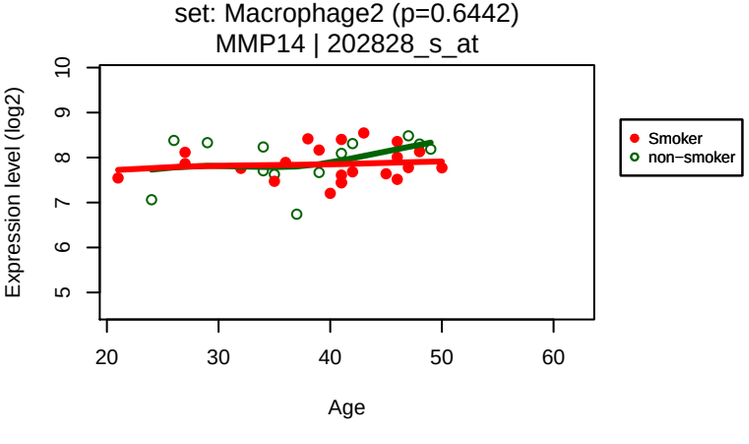

set: SmallAirway2 (p=0.1888)
MMP14 | 202828_s_at
Expression level (log2)
Age
Smoker
non−smoker

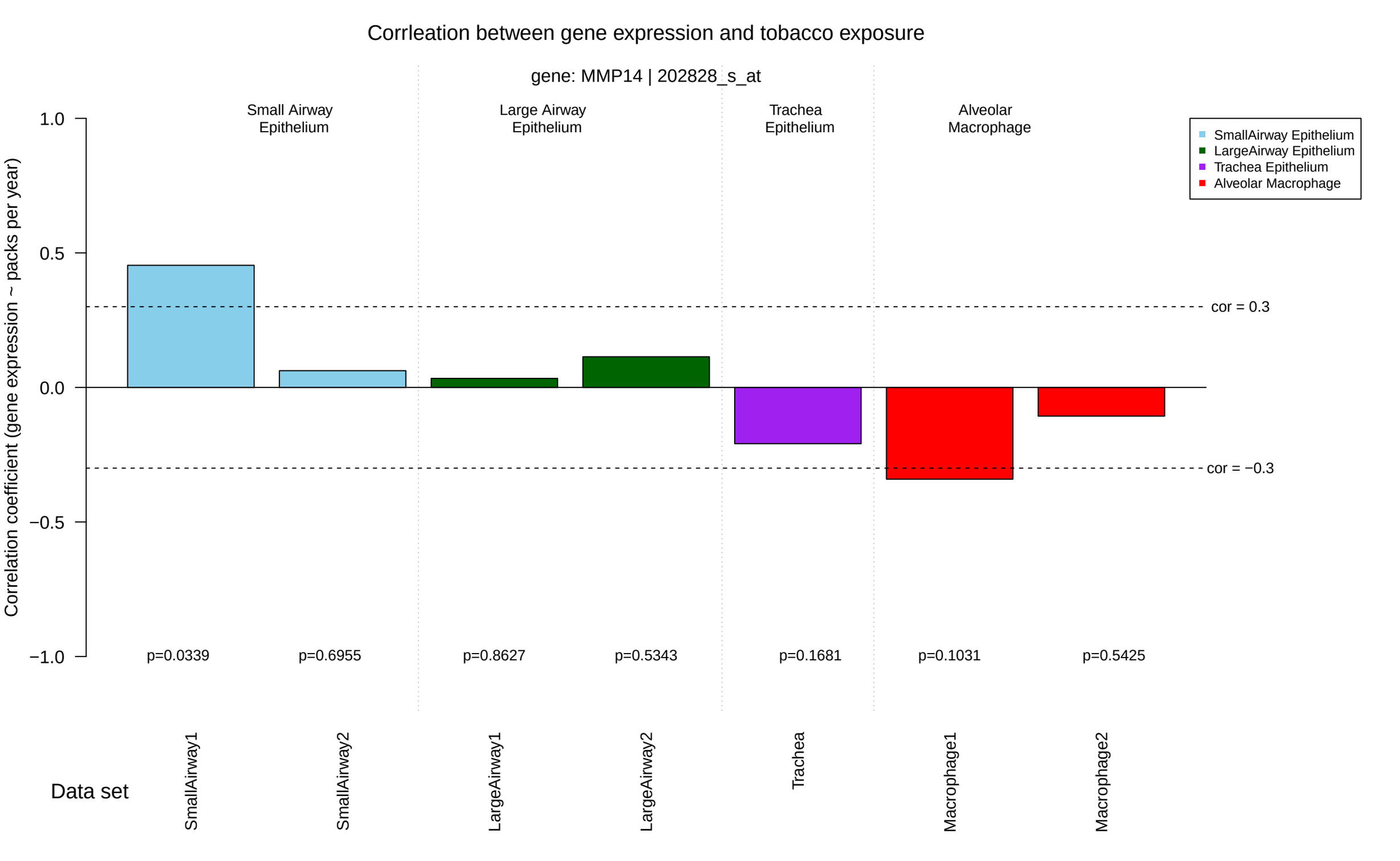

set: LargeAirway1 (p=0.2091)
MMP14 | 202828_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.265)
MMP14 | 202828_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4234)
MMP14 | 202828_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0332)
MMP14 | 202828_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6442)
MMP14 | 202828_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP14 | 202828_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0339
p=0.6955
p=0.8627
p=0.5343
p=0.1681
p=0.1031
p=0.5425
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2